COMSATS University Islamabad

Department of Physics

EEE351 – Principles of Communication Systems (Fall 2026)

SEMESTER PROJECT PROPOSAL

Design and Simulation of a Complete Digital Communication Link with Error Correction

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1. Group Members

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2. Introduction

A digital communication system converts an analog message into bits, transmits them over a noisy channel, and reconstructs the message at the receiver. Each stage, from sampling to error correction, is studied separately in EEE351. This project brings these stages together into one working end-to-end simulation so that the effect of each design choice on the overall system can be observed and measured.

3. Problem Statement

Theoretical results such as the sampling theorem, quantization noise, and bit error rate are difficult to appreciate without seeing them in action. The project aims to build a software model of a complete digital link and to study how noise, quantization, modulation scheme, and channel coding affect the quality of the received signal.

4. Objectives

- Implement a full digital communication chain in Python: source, sampling, PCM, line coding, digital modulation, AWGN channel, demodulation, decoding, and reconstruction.
- Demonstrate the sampling theorem and aliasing, and measure SNR against the number of quantization bits for uniform and μ-law quantization.
- Compare the BER performance of BPSK and QPSK (and FSK if time permits) over an AWGN channel against theoretical curves.
- Show the improvement in BER obtained by adding Hamming error-correcting code.
- Transmit an image through the noisy link and show the received image with and without error correction.
- Develop an interactive Streamlit interface to control the system parameters and view the results live.

5. Course Topics Covered

Project Block	Related Course Topic
Sampling and reconstruction	Sampling theorem, interpolation, aliasing
Quantization and PCM	Uniform and non-uniform quantization, A-law/μ-law, SNR vs bits
Line coding	Baseband digital data transmission, line coding schemes
Digital modulation	ASK, FSK, PSK, QAM and digital modulation schemes
Noisy channel	SNR, bandwidth, Shannon capacity and trade-offs
Error correction	Coding and error-correcting codes
Image transmission	Digital data transmission and the effect of channel errors on received data

6. Methodology

The system will be developed block by block. Each block is written as an independent function and tested on its own before being connected to the next one:

- Source and sampling: a short audio clip is sampled at different rates to demonstrate aliasing.
- PCM: the samples are quantized (uniform and μ-law) and converted into a bit stream.
- Line coding: the bit stream is converted to NRZ and Manchester waveforms for comparison.
- Modulation: bits are mapped onto BPSK and QPSK symbols.
- Channel: additive white Gaussian noise is added at varying SNR values.
- Receiver: demodulation, decision making, Hamming decoding, and reconstruction of the audio.
- Image transmission: a grayscale image is converted to bits, sent through the same link, and reconstructed to compare quality with and without Hamming coding.
- Analysis: BER vs SNR curves, constellation diagrams, and SNR vs quantization bits plots.
- Interface: a Streamlit web interface built on top of the tested functions, with controls for SNR, modulation type, and coding on/off, and live plots, audio playback, and image display.

7. Tools and Technologies

- Python 3 with NumPy, SciPy, and Matplotlib for simulation and plotting.
- Streamlit for the interactive user interface, and Pillow for image handling.
- Jupyter Notebook for documenting and presenting results.
- Git and GitHub for version control and team collaboration.

8. Expected Outcomes and Deliverables

- A working simulation of a digital communication link with reconstructed audio at the output.

- Plots of BER vs SNR (with and without error correction), constellation diagrams, and quantization SNR curves.
- A visual demonstration of an image transmitted over a noisy channel, received with and without error correction.
- An interactive interface for live demonstration of the complete system.
- Source code, a final report, and a demonstration at the time of the lab evaluation.

9. Applications

- Teaching and learning tool that shows how each stage of a digital communication system affects the received signal.
- Basis for studying real systems such as digital audio (PCM), mobile communication, Wi-Fi, and satellite links.
- Quick testing and comparison of modulation and coding schemes before hardware implementation.
- Foundation for research and projects in software-defined radio and wireless communication.

10. Future Upgrades

- Add higher-order modulation (16-QAM, 64-QAM) and compare spectral efficiency against BER.
- Add stronger error-correcting codes such as convolutional codes with Viterbi decoding, or Reed-Solomon codes.
- Add fading channel models (Rayleigh) and multipath effects alongside AWGN.
- Add pulse shaping (raised cosine) and inter-symbol interference analysis with an equalizer.
- Add a machine learning module that identifies the modulation type of a noisy signal.
- Transmit text and other file types through the link in addition to audio and images.
- Test the link with a real SDR device (RTL-SDR or similar).

11. References

- B. P. Lathi and Z. Ding, Modern Digital and Analog Communication Systems.
- L. W. Couch, Digital and Analog Communication Systems.
- NumPy, SciPy, and Matplotlib official documentation.